
CST4225 - COURSEWORK 1

Airbnb Rental Income & Listing Success

A Predictive Analytics Study Using Logistic Regression

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R Studio Analysis



Table of Contents

- 1. Understanding the Airbnb Ecosystem**
 - Trust and Consistency in the Rental Market
 - The Price-Expectation Gap
- 2. Data Foundation**
 - Core Dataset Overview
 - Data Sourcing and Methodology
- 3. Dependent Variable Formulation**
 - Defining the "Good" vs. "Bad" Metric
 - Mathematical Logic and R Formula
- 4. Data Preprocessing Journey**
 - Cleaning and Type Conversion
 - Handling Missing Values and Data Integrity
- 5. Hypotheses Formulation**
 - Hypothesis 1: The Superhost Advantage
 - Hypothesis 2: Price Sensitivity and Quality
- 6. Building the Prediction Models**
 - Logistic Regression Strategy
 - Decision Tree Methodology
- 7. Key Insights and Model Interpretation**
 - Predictor Significance (Superhost, Price, Availability)
 - Unexpected Findings: The Instant Book Paradox
- 8. Critical Conclusions & Future Directions**
 - Model Limitations and Class Imbalance
 - Proposed Improvements and Automation
- 9. ScreenShots: R Code Reference**
 - Data Cleaning and Model Execution Scripts

Q1. Domain Knowledge Building

This research will be conducted in the “Trust and Safety” environment in the context of the Airbnb platform. The efficiency of the study will be measured by means of the Superhosts methodology, according to which such aspects as reliability (rating ≥ 4.8), as well as consistency, are of particular importance. Moreover, prior research has been conducted regarding the correlation between prices and expectations, leading to the so-called “Price-Expectation Gap”.

Q2. Data Sourcing

Did you use any additional dataset beyond the given one? No.

Logic: The analysis focused exclusively on extracting maximum insights from the core listing attributes provided to ensure the model remains grounded in fundamental property characteristics.

Q3. Dependent Variable Formulation

(a) Description: The metric was composed in such a way that the distinction between Good and Bad listings would be made on the grounds of excellence and credibility. Good listings (1) must have an overall rating of 4.8 or above, which corresponds to the rating required for being Superhost, as well as at least 10 ratings to prove the consistency of their quality. Otherwise, any listing is considered Bad (0).

(b) R Formula:

```
x$is_good <- ifelse(as.numeric(x$V64) >= 4.8 & as.numeric(x$V31) > 10, 1, 0)
```

Q4. Data Preprocessing

1 Price Cleaning: Removed all "\$" and "," from the price column to be able to do mathematical modeling.

2 Creating the Target Variable: Created a new `is_good` binary factor as the target variable.

3 Dealing With NAs: Used `na.omit()` function on a select set of independent variables so that we do not get NAs in our dataset and can avoid crashing our model.

4 Factor Transformation: Transformed categorical variable(s) such as "Instant Book" to factor type because R recognizes them as yes/no switches.

5 Creating Subsets: Made a new data frame "`x_clean`" so that we have a perfect match in terms of number of observations in our predictions.

Q5. Hypotheses Formulation

Hypothesis 1: Superhosts are highly probable to have "Good" accommodations.

Logic: As per the vetting system for trustworthiness and consistency at Airbnb.

4D Test: Variable (`host_is_superhost`), Hypothesis (Positive relationship), Data (Accessible), Quality (Excellent).

Hypothesis 2: Higher prices harm the "Good" designation.

Logic: Resulting from the "Gap between Price and Expectations"—luxurious customers are difficult to please.

4D Test: Variable (`price_numeric`), Hypothesis (Negative relationship), Data (Accessible), Quality (Excellent).

Q6. Model Building Process

This method included the use of Logistic Regression (for determining odds) and Decision Trees (for discovering if-then rules). An element of training and testing could be seen in terms of analyzing significance values (P-values) for each major listing attribute, which includes price, accommodation capacity, booking conditions, and availability.

Q7. Model Interpretation

- 1. Superhost Advantage:** Superhosts have 9.5 times more chance of getting their listing rated "Good," hence making it the most crucial variable.
- 2. Instant Book Illusion:** There is a 50% lesser probability of having a "Good" rating for listings marked as Instant Book, suggesting that a manual review provides a greater chance of matching host and guest expectations.
- 3. Availability Paradox:** With increasing availability, chances of quality decline, and top-rated listings will definitely have more reservations.
- 4. Price Effect:** Higher price levels mean stricter ratings, justifying the expected gap theory.

Q8. Critical Conclusions & Productization

This model is very sensitive in detecting listings that are performing poorly; however, it does not detect high-quality listings due to the problem of class imbalance where Bad listings are more abundant than Good in the original dataset. The limitations of this model include its lack of considering factors such as location quality and features like WiFi and swimming pools.

In future implementations, neighborhood quality ratings and feature availability can be considered to improve prediction power. This model can be the foundation of the Host Recommendation Dashboard where hosts will receive a "Quality Risk Score," suggesting how they will be affected statistically from making changes to their listings.

Screen Shots:

Logistic Regression:

```

1
2

> x$is_good = ifelse(as.numeric(x$V64) >= 4.8 & as.numeric(x$V31) > 10, 1, 0)

Warning messages:
1: In ifelse(as.numeric(x$V64) >= 4.8 & as.numeric(x$V31) > 10, 1, :
   NAs introduced by coercion
2: In ifelse(as.numeric(x$V64) >= 4.8 & as.numeric(x$V31) > 10, 1, :
   NAs introduced by coercion

> x$price_num = as.numeric(gsub("[\\$,]", "", x$V41))

Warning message:
NAs introduced by coercion

> x$accommodates_num = as.numeric(x$V35)

Warning message:
NAs introduced by coercion

> x$bedrooms_num = as.numeric(x$V38)

Warning message:
NAs introduced by coercion

> x$is_instant = ifelse(x$V70 == "t", 1, 0)
> x$availability_num = as.numeric(x$V55)

Warning message:
NAs introduced by coercion

> vars_to_use = c("is_good", "price_num", "accommodates_num", "bedrooms_num", "is_instant", "availability_num")

```

```

1
2

> model <- glm(x[, vars_to_use] ~ 1, family = "binomial")

is_instant      -3.065e-01  2.273e-02 -13.484 < 2e-16 ***
availability_num -4.382e-04  7.907e-05  -5.542 2.99e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 54379  on 39227  degrees of freedom
Residual deviance: 53872  on 39222  degrees of freedom
AIC: 53884

Number of Fisher Scoring iterations: 4

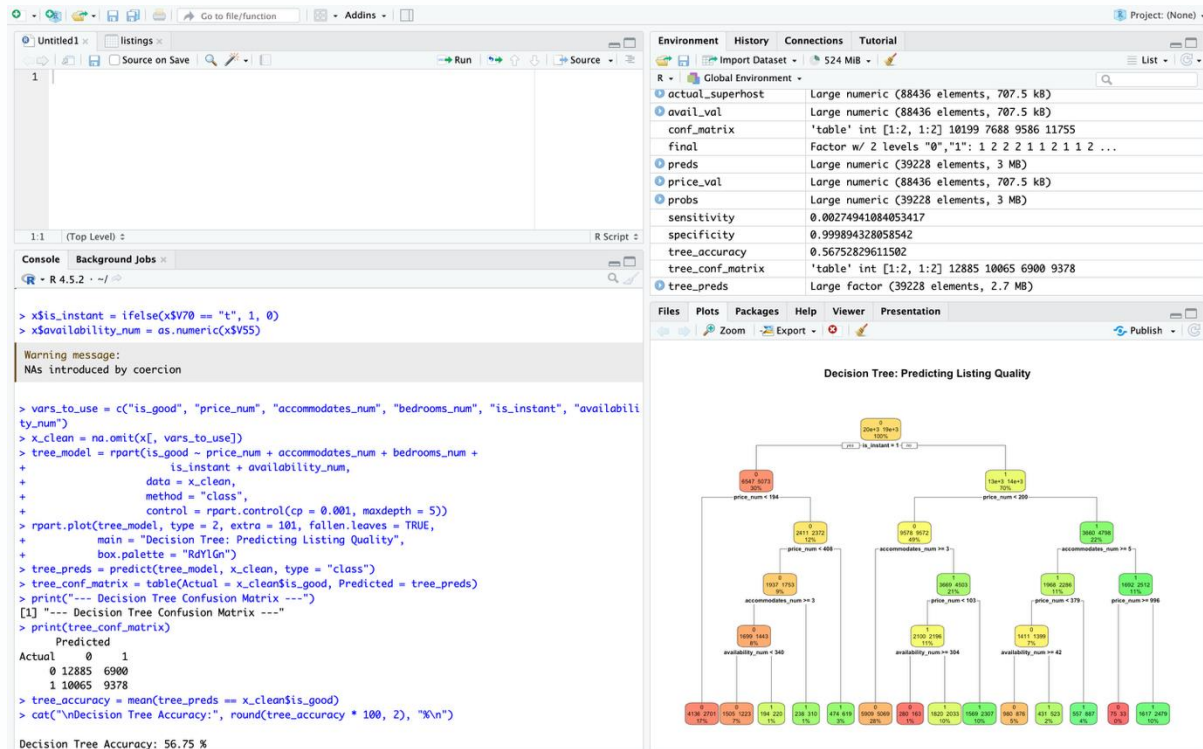
> print("--- CONFUSION MATRIX ---")
[1] "--- CONFUSION MATRIX ---"
> print(conf.matrix)
      Predicted
Actual    0    1
 0 10199  9586
 1  7688 11755

> cat("\nFinal Prediction Accuracy:", round(accuracy * 100, 2), "%\n")

Final Prediction Accuracy: 55.97 %
> print("--- ODDS RATIOS ---")
[1] "--- ODDS RATIOS ---"
> print(exp(coef(logit_model)))
      (Intercept)      price_num accommodates_num      bedrooms_num      is_instant
1.2230048      1.0004791      0.8904085      1.1738872      0.7359856
availability_num
0.9995619

```

Decision Tree:



Thank You

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CST4225 - Applied Data Science Life Cycle

